

Lab-II problem-2

Date: 10-02-2026

The data on weight, height and chest circumference of 15 UG students are given below.

Weight(kg)	Height(cm)	Chest circumference(cm)
60	155	36
61	158	45
54	155	39
49	146	34
63	165	46
69	170	49
67	170	46
50	150	39
57	153	45
55	151	40
59	155	41
67	171	47
63	161	41
49	150	39
50	152	40

Find the sample mean vector $\bar{X}$, population variance-covariance matrix Cov(X) and population correlation matrix Cor (X) of the above data.

Solution:-

Source: R Studio

Program:

```
weight = c(60,61,54,49,63,69,67,50,57,55,59,67,63,49,50)
```

```
weight
```

```
height = c(155,158,155,146,165,170,170,150,153,151,155,171,161,150,152)
```

```
height
```

```
chest = c(36,45,39,34,46,49,46,39,45,40,41,47,41,39,40)
```

```
chest
```

```
# Combine into matrix
```

```
X = cbind(weight, height, chest)
```

```
X
```

```
x_bar = colMeans(X)
```

```
x_bar
```

```
cov_X = cov(X)
```

```
cov_X
```

```
cor_X = cor(X)
```

```
cor_X
```

Output:-

```
> weight = c(60,61,54,49,63,69,67,50,57,55,59,67,63,49,50)
```

```
> weight
```

```
[1] 60 61 54 49 63 69 67 50 57 55 59 67 63 49 50
```

```
> height = c(155,158,155,146,165,170,170,150,153,151,155,171,161,150,152)
```

```
> height
```

```
[1] 155 158 155 146 165 170 170 150 153 151 155 171 161 150 152
```

```
> chest = c(36,45,39,34,46,49,46,39,45,40,41,47,41,39,40)
```

```
> chest
```

```

[1] 36 45 39 34 46 49 46 39 45 40 41 47 41 39 40
> X = cbind(weight, height, chest)
> X
      weight height chest
[1,]      60     155    36
[2,]      61     158    45
[3,]      54     155    39
[4,]      49     146    34
[5,]      63     165    46
[6,]      69     170    49
[7,]      67     170    46
[8,]      50     150    39
[9,]      57     153    45
[10,]     55     151    40
[11,]     59     155    41
[12,]     67     171    47
[13,]     63     161    41
[14,]     49     150    39
[15,]     50     152    40
> x_bar = colMeans(X)
> x_bar
      weight      height      chest
58.2000 157.4667 41.8000
> cov_X = cov(X)
> cov_X
      weight      height      chest
weight 47.31429 52.11429 23.11429
height 52.11429 65.69524 28.95714
chest 23.11429 28.95714 18.60000
> cor_X = cor(X)
> cor_X
      weight      height      chest
weight 1.0000000 0.9347462 0.7791622
height 0.9347462 1.0000000 0.8283851
chest 0.7791622 0.8283851 1.0000000

>

```

Conclusion:

From the analysis we obtained sample mean vector

```

      weight      height      chest
58.2000 157.4667 41.8000

```

Population variance covariance matrix cov(x)

```

      weight      height      chest
weight 47.31429 52.11429 23.11429
height 52.11429 65.69524 28.95714
chest 23.11429 28.95714 18.60000

```

Population correlation matrix cor(x)

```

      weight      height      chest
weight 1.0000000 0.9347462 0.7791622
height 0.9347462 1.0000000 0.8283851
chest 0.7791622 0.8283851 1.0000000

```